

4. Address Space

4.1 Address Space

This MCU has a 4-Gbyte address space, consisting of the range of addresses from 0000 0000h to FFFF FFFFh. That is, linear access to an address space of up to 4 Gbytes is possible, and this contains both program and data areas.

Figure 4.1 shows the memory maps in the respective operating modes. Accessible areas will differ according to the operating mode and states of control bits.

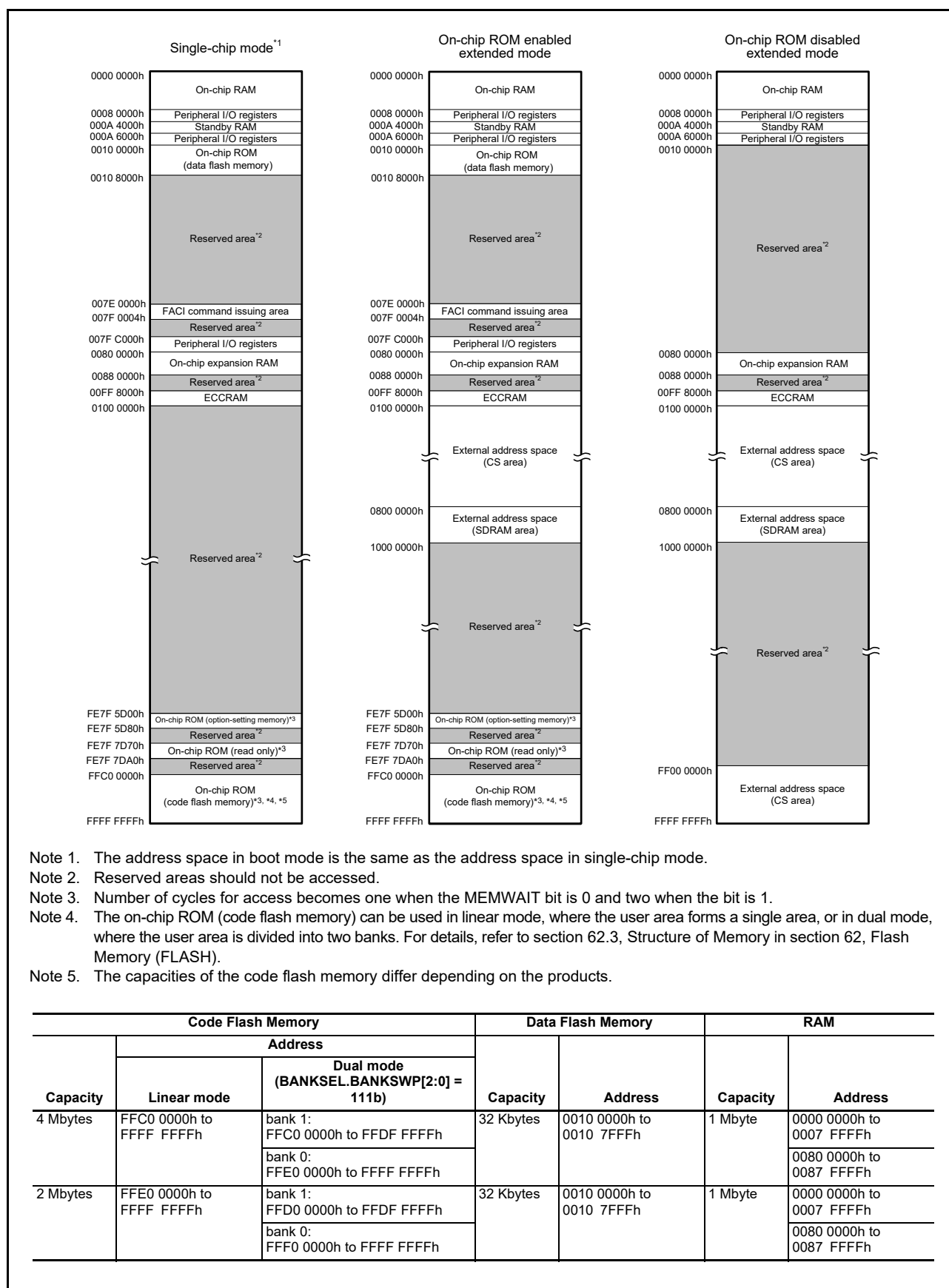
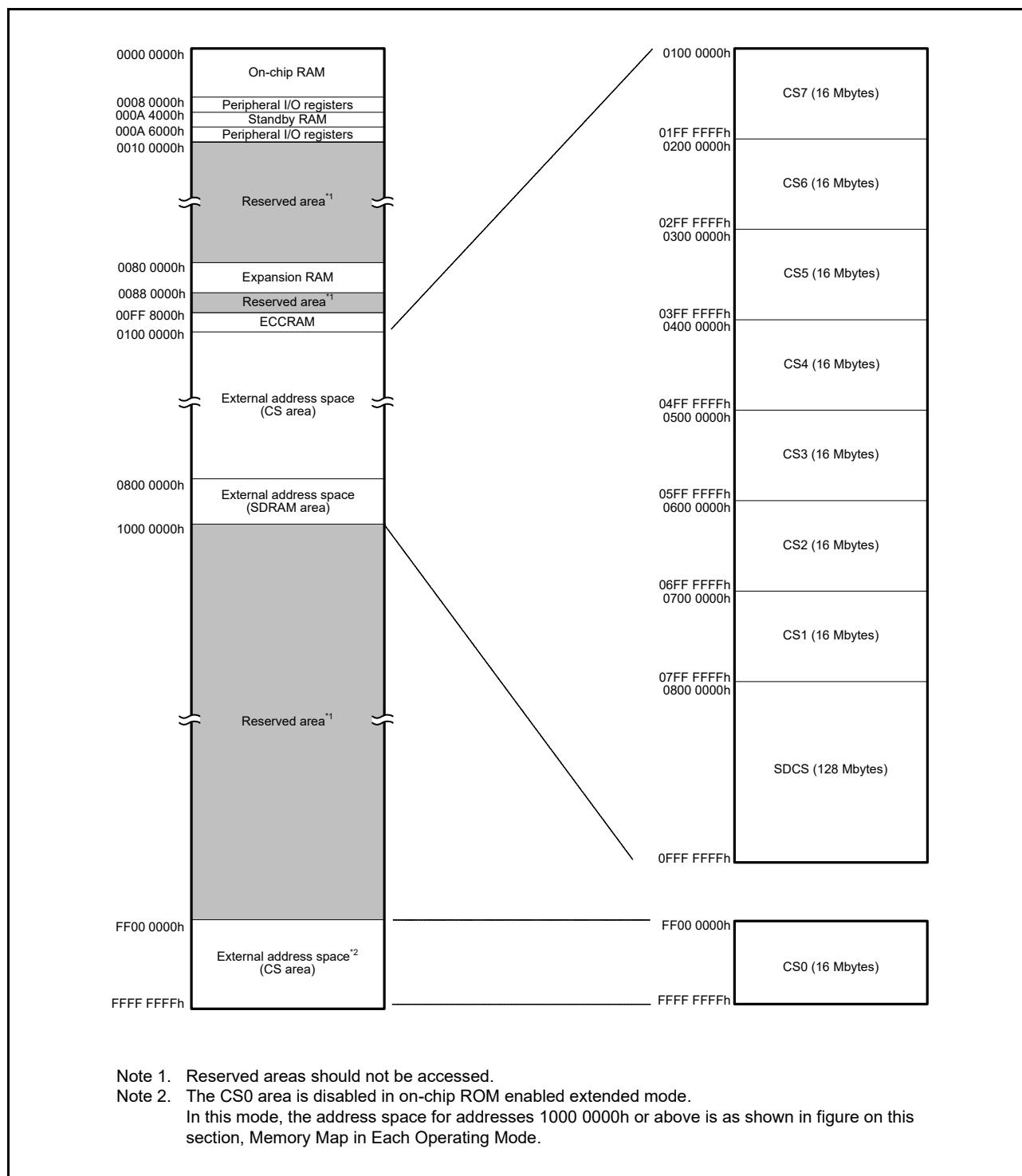


Figure 4.1 Memory Map in Each Operating Mode

4.2 External Address Space

The external address space is divided into CS areas (CS0 to CS7) and SDRAM area (SDCS). The CS areas are divided into up to eight areas (CS0 to CS7), each corresponding to the CSn# signal output from a CSn# (n = 0 to 7) pin.

Figure 4.2 shows the address ranges corresponding to the individual CS areas (CS0 to CS7) and SDRAM area (SDCS) in on-chip ROM disabled extended mode.



**Figure 4.2 Correspondence between External Address Spaces and CS Areas
(In On-Chip ROM Disabled Extended Mode)**